

Multi-trait genome-wide analysis identified risk loci and candidate drugs for heart failure

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Summary

Heart failure (HF) is a common cardiovascular syndrome that poses significant morbidity and mortality risks. While genome-wide association studies reporting on HF abound, its genetic etiology remains poorly elucidated, primarily due to its inherent polygenic nature. Furthermore, these genetic insights have not been fully leveraged to develop effective primary treatment strategies for HF. In this study, we conducted a large-scale integrated multi-trait analysis using European ancestry genome-wide association study summary statistics of coronary artery disease and HF, involving nearly 2 million samples to identify risk loci associated with HF. Seventy-two loci were identified for HF using MTAG, of which 58 were supported in the replication phase. Transcriptome association analysis revealed 215 HF risk genes, including *EDNRA* and *FURIN*. Pathway enrichment analysis of risk genes revealed their enrichment in pathways closely related to HF, such as response to endogenous stimulus (adjusted $p = 8.83 \times 10^{-3}$), phosphate-containing compound metabolic process (adjusted $p = 1.91 \times 10^{-2}$), myofibroblast differentiation (adjusted $p = 4.26 \times 10^{-2}$), and regulation of muscle adaptation (adjusted $p = 4.96 \times 10^{-2}$). Single-cell analysis indicated significant enrichments of these genes in smooth muscle cells, fibroblasts of cardiac tissue, and cardiac endothelial cells. Additionally, our analysis of HF risk genes identified 81 potential drugs for further pharmacological evaluation. These findings provide insights into the genetic determinants of HF, highlighting MTAG-identified genetic loci as potential interventional targets for HF treatment, with significant implications for public health and clinical practice.

Introduction

Heart failure (HF) is a common cardiovascular syndrome that causes symptoms such as shortness of breath, volume overload, and other functional limitations, resulting from structural or functional impairment of ventricular filling or ejection of blood.¹ It poses a significant global health challenge with high prevalence, morbidity, mortality, hospitalization rates, and healthcare costs.² Understanding the genetic etiology of HF is crucial, as it can provide insights into the molecular mechanisms underlying the disease, identify therapeutic targets, and improve risk stratification through genetic markers.

However, the polygenic nature of HF, with numerous small-effect genetic variants, poses significant challenges to fully elucidating its genetic basis. Compared with other cardiovascular diseases, fewer HF-associated loci have been identified, due to limited statistical power in genome-wide association studies (GWASs) and smaller sample sizes.^{3–5} Additionally, HF exhibits moderate heritability (approximately 26%)⁶ compared with other cardiovascular traits like coronary artery disease (CAD), further constraining genetic discovery using conventional approaches. Despite this moderate heritability, HF likely

harbors numerous causal variants with small individual effects that collectively account for its heritable component. The genetic architecture of HF is further complicated by heterogeneity across subtypes (e.g., HF with preserved ejection fraction versus reduced ejection fraction) and by the modest availability of diverse population datasets. Linkage disequilibrium further complicates the prioritization of causal variants and genes.

MTAG (multi-trait analysis of GWAS) emerged as a powerful tool for analyzing complex diseases like HF. By leveraging genetic correlations among HF and related traits such as CAD and hypertension, MTAG enhances statistical power and facilitates the discovery of loci that may not achieve significance in single-trait GWAS. This methodology has already demonstrated success in identifying new loci for cardioembolic stroke,⁷ primary sclerosing cholangitis,⁸ and Lewy body dementia.⁹ For HF research, MTAG's ability to distinguish shared from trait-specific genetic effects is especially critical, given the disease's strong genetic correlation with CAD ($r_g = 0.67$).⁵ Longitudinal studies corroborate this interplay, showing that CAD patients face elevated HF risk.¹⁰ The shared genetic architecture between HF and CAD provides a strategic entry point for MTAG-driven discovery.

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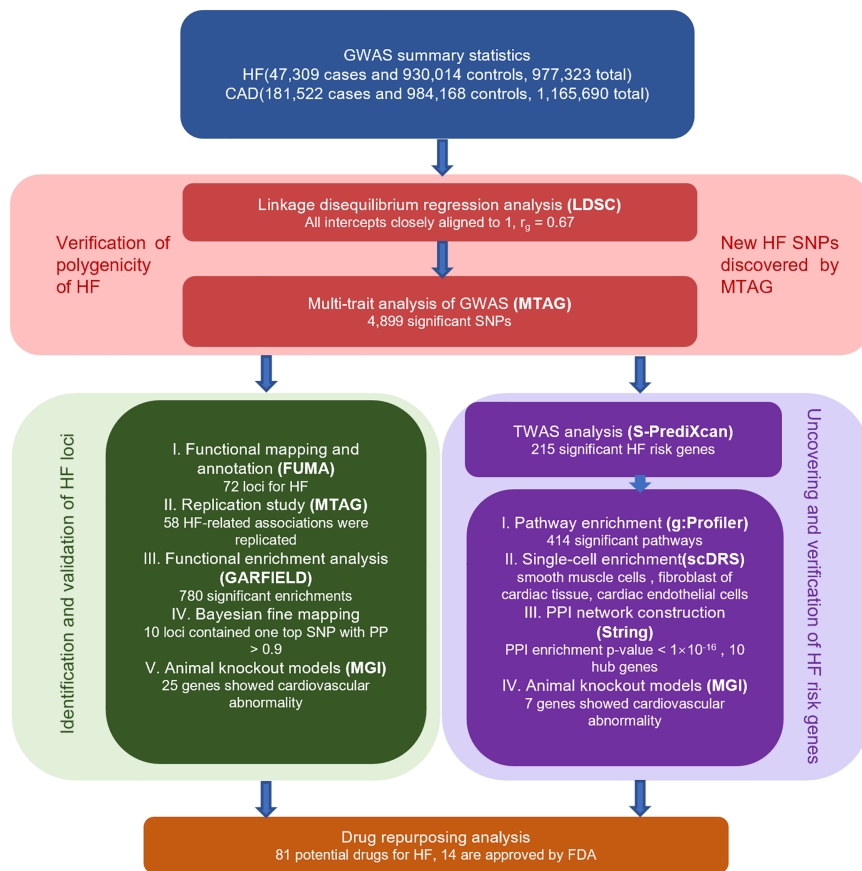


Figure 1. Overall study design

Employing the largest available GWAS summary statistics for HF encompassing 47,309 cases and 930,014 controls, as well as CAD with 181,522 cases and 984,168 controls, we explored the polygenicity and genetic correlation between them using LDSC, followed by MTAG to identify significant SNPs. Functional annotation at the SNP level was performed to identify risk loci paired with MTAG on replication phase to confirm the MTAG-identified risk loci. Additionally, TWAS was conducted to uncover HF risk genes. Subsequently, four gene-based analyses (pathway enrichment, single-cell enrichment, protein-protein interaction network, and animal knockout models) were performed to confirm the role of TWAS risk genes in the development of HF. Finally, drug repurposing was employed to identify potential therapeutic drugs for HF.

conducting signature mapping. Specifically, this involves correlating expression level of genes associated with the disease with the pharmacological effects of compounds acting on these genes.^{23–26}

In this study, we combined MTAG with complementary statistical methods to comprehensively investigate the genetic basis of HF (Figure 1).

Beyond MTAG, advanced computational methods such as transcriptome-wide association studies (TWAS),^{11–13} linkage disequilibrium score regression (LDSC),^{14,15} and functional annotation tools (e.g., functional mapping and annotation [FUMA]¹⁶) further refine causal gene prioritization. While TWASs integrate multi-omics data to map variants to gene expression and LDSC quantifies genetic correlations, these tools are often applied in isolation rather than as a unified framework. The growing availability of large-scale GWAS datasets—notably from the GWAS Catalog¹⁷ and FinnGen¹⁸—now addresses historical limitations of sample size and population diversity. These resources enable systematic dissection of HF’s genetic architecture.

Moreover, drug repurposing has emerged as an efficient approach to translate approved drugs for new indications based on GWAS-identified risk genes. Currently, although several medications exist for treating HF, most only alleviate symptoms and do not reverse the condition. For HF with preserved ejection fraction, only SGLT2 inhibitors have demonstrated significant reductions in hospitalization and cardiovascular death for patients.¹⁹ Consequently, there is an urgent need to discover therapeutic drugs for HF, and drug repurposing offers a promising avenue. In recent years, mapping of genome-wide significant GWAS loci to specific drug-targeted genes, thus enabling drug reuse, has become a critical approach.^{20–22} Another approach involves using TWASs^{11,13} to obtain gene information associated with the phenotype and

conducting signature mapping. Specifically, this involves correlating expression level of genes associated with the disease with the pharmacological effects of compounds acting on these genes.^{23–26} In this study, we combined MTAG with complementary statistical methods to comprehensively investigate the genetic basis of HF (Figure 1). Leveraging the most extensive European GWAS summary statistics datasets for HF and CAD, we identified 99 HF-associated loci, including 72 MTAG-identified loci, 58 of which were supported in a replication phase. Further integration with TWAS uncovered 215 risk genes linked to HF, such as *EDNRA* (OMIM: 131243) and *FURIN* (OMIM: 136950). Pathway enrichment analysis revealed that these genes are involved in critical biological processes, including response to endogenous stimulus, phosphate metabolism, myofibroblast differentiation, and regulation of muscle adaptation. Single-cell analysis further localized these genes to smooth muscle cells, cardiac fibroblasts, and endothelial cells, emphasizing their functional relevance. Finally, drug repurposing analysis using MTAG and TWAS findings suggested 81 potential therapeutic candidates for HF, including 14 Food and Drug Administration (FDA)-approved drugs, offering promising avenues for clinical application and future research.

Materials and methods

Study design, data source, and quality control

The complete workflow of this study is illustrated in Figure 1. We obtained the largest GWAS summary statistics of HF (47,309 cases and 930,014 controls) and CAD (181,522 cases and 984,168 controls) from the GWAS Catalog (<https://www.ebi.ac.uk/gwas/>)^{5,17,27} for the discovery phase. Additionally, we also

obtained GWAS summary statistics for HF (19,350 cases and 288,996 controls) and for CAD (33,628 cases and 275,526 controls) from the FinnGen¹⁸ repository <https://finngen.gitbook.io/documentation/v/r7/>,¹⁸ serving as an independent replication phase. These studies were all conducted on individuals of European ancestry and underwent stringent quality control procedures, as previously described.^{5,18,27}

In the CAD summary statistics, base pair positions were converted to rs ID, employing the Genome Reference Consortium Human Reference 37 or Hg19 as deposited in the UCSC Genome Browser ([/goldenpath/hg19/database](https://goldenpath/hg19/database)). Furthermore, we excluded the major histocompatibility complex region (chromosome 6, 26–34 Mb) from our analyses. Single-nucleotide polymorphisms (SNPs) with minor allele frequency less than 0.01 were filtered out, and the analysis was restricted to biallelic SNPs. SNPs with duplicated or missing rs IDs were removed from each GWAS summary dataset for subsequent analyses. A detailed description of all GWAS studies used in this research can be found in [Table S1](#).

LDSC regression analysis

To estimate the polygenicity of HF and the genetic correlation between HF and CAD, we used LDSC.^{14,15} We first performed single-trait LDSC analysis on the GWAS summary statistics of HF and CAD to obtain several key parameters, including the mean χ^2 statistic, genomic inflation factor (λ_{GC}), and intercept.

The mean χ^2 statistic measures the association between genetic variants and a specific trait of interest in the GWAS summary statistics. The traits with a mean χ^2 statistic <1.1 were excluded from further analysis. The factor λ_{GC} evaluates the presence of genetic inflation, which refers to excess test statistics over the expected null distribution. A λ_{GC} value close to 1 indicates little evidence of systematic bias, whereas values substantially greater than 1 may arise from population stratification, cryptic relatedness, or polygenic effects. The intercept obtained from LDSC analysis (intercept minus 1), assesses the contribution of confounding biases to the inflation of test statistics. A value close to 1 suggests that polygenic effects primarily drive inflation.

Subsequently, we applied pairwise LDSC analysis to investigate the genetic correlations between HF and CAD. We used the pre-computed linkage disequilibrium scores of European ancestry downloaded from the 1000 Genomes Project Phase 3 (<https://alkesgroup.broadinstitute.org/LDSCORE/>).²⁸ Since that low imputation quality can potentially diminish test statistics, we focused on well-imputed HapMap3 SNPs.

Multi-trait meta-analysis with MTAG

To boost the power of trait-associated SNP discovery, we used MTAG v.1.0.8.²⁹ MTAG is a robust method designed to enhance trait-specific power by leveraging genetic correlations among multiple genetically related traits.²⁹ As input, it takes the GWAS summary statistics of correlated traits and incorporates the LDSC method to estimate the genetic covariance matrix. Subsequently, it generates weights and employs generalized inverse variance weighting to estimate trait-specific effects for a shared set of SNPs. In our analysis, MTAG used SNP information (SNP ID, p value, and beta) from both HF and CAD as input and generated SNP-level statistics for each trait separately (i.e., MTAG_{HF} and MTAG_{CAD}). We then focused the downstream analyses on MTAG_{HF}.

To replicate HF-associated loci identified from MTAG, we also conducted MTAG on HF and CAD from the FinnGen repository,¹⁸ independent of those in discovery phase. In our analysis,

the summary statistics derived from single-trait GWAS were denoted as GWAS_{HF} and GWAS_{CAD}. Additionally, results from the replication phase were denoted as MTAG_{HF_R} and MTAG_{CAD_R}. The genome-wide significance level for MTAG_{HF} was set at $P_{MTAG_HF} < 5 \times 10^{-8}$.

Functional annotation by FUMA

We used the FUMA platform v.1.5.4¹⁶ to investigate the functional implications of HF-associated SNPs. Using its default settings, we annotated significant SNPs of MTAG_{HF}, using the European ancestry data from the 1000 Genomes Project Phase 3 as a reference. We defined independent significant SNPs as those with $P_{MTAG_HF} < 5 \times 10^{-8}$ and independent from each other at $r^2 < 0.6$. Lead SNPs are independently significant SNPs that are independent from each other at $r^2 < 0.1$. Candidate SNPs are those in high LD with an independent significant SNP ($r^2 \geq 0.6$). Genomic risk loci are identified by merging linkage disequilibrium blocks of independent significant SNPs in close proximity (<250 kb). The top lead SNP is recognized as the SNP with the lowest P_{MTAG_HF} ¹⁶ within a locus.

Through this FUMA annotation, we defined three key gene sets: MTAG_{HF} genes refer to genes mapped from significant SNPs ($P_{MTAG_HF} < 5 \times 10^{-8}$) identified through MTAG analysis; GWAS_{HF} genes refer to genes mapped from significant SNPs ($P_{GWAS_HF} < 5 \times 10^{-8}$) in the original GWAS_{HF} analysis; and GWAS_{CAD} genes refer to genes mapped from significant SNPs that reached the genome-wide significance threshold in the original GWAS_{CAD} analysis. All gene sets were mapped from significant SNPs using the FUMA annotation tool, including genes located at or near these SNPs.

To gain further insight into the functional annotations of independent significant SNPs and their LD proxies, we employed tools including ANNOVAR categories,³⁰ combined annotation-dependent depletion (CADD) scores,³¹ and Regulome DB scores³² with their default parameters. We applied the same procedure to genome-wide significant SNPs from GWAS_{HF} and GWAS_{CAD} for comparative purposes.

Bayesian fine-mapping analysis

To prioritize causal SNPs for each locus, we conducted Bayesian fine-mapping analysis to analyze risk loci determined in the FUMA analysis using the *finemap.abf* function of the *coloc* v5 R package (<https://chr1swallace.github.io/coloc/>).³³ This function leverages Bayesian methods to calculate the posterior probability for each SNP being a causal variant within the corresponding risk locus.³³ SNPs were incrementally included into the set based on their descending posterior probabilities, up to the point where the cumulative posterior probability reached 0.90, resulting in a 90% credible SNP set for each locus.

Functional enrichment analysis

To investigate the functional enrichments of significant SNPs of MTAG_{HF}, we used GWAS analysis of regulatory or functional information enrichment with LD correction (GARFIELD).³⁴ GARFIELD utilizes GWAS summary statistics data, and regulatory functional annotation data, including genic annotations, histone modifications, transcription factor binding sites, and chromatin segmentation states, across cell types and tissues.³⁴ Specifically, GARFIELD applies LD pruning based on LD and distance information to select independent SNPs, and annotates SNPs with regulatory information if they, or highly correlated SNPs with them,

overlap with such regulatory features. Logistic regression is then performed, in which SNPs are coded as binary outcomes according to whether their GWAS p value passes a predefined threshold, and functional annotations and covariates (such as LD proxies and distance to transcription start site) are included as predictors. Odds ratios (OR) and enrichment p values are subsequently computed. We determined the enrichment to be significant at Bonferroni-corrected $p < 4.98 \times 10^{-5}$ (0.05/1005), where 1,005 represents the number of feature annotations. We also applied GARFIELD to GWAS_{HF} for comparison.

Transcriptome-wide association analysis

To fully harness the information from SNPs for prioritizing risk genes associated with HF, we used the S-PrediXcan method¹³ combined with the joint transcriptome imputation (JTI) model³⁵ to perform TWAS analysis on MTAG_{HF}. S-PrediXcan integrates summary-level GWAS data with gene expression prediction models, estimating the association between gene expression and HF.¹³ To strengthen the accuracy and robustness of S-PrediXcan analysis, we applied the JTI model. This model combines information from multiple tissue-specific gene expression prediction models, enabling more effective integration of gene expression data across tissues.³⁵ We used JTI models derived from the genotype-tissue expression project version 8 transcriptome data,³⁶ for eight different tissues associated with the heart. These tissues include subcutaneous adipose tissue, visceral omental adipose tissue, atrial appendage, left ventricle, kidney cortex, liver, lung, and whole blood. We applied the Bonferroni correction for multiple tests on each tissue. From these analyses, we defined two sets of risk genes: TWAS_{MTAG} risk genes refer to genes showing significant association with HF in at least one tissue through TWAS analysis of MTAG_{HF} results after Bonferroni correction; and TWAS_{GWAS} risk genes refer to genes showing statistically significant association with HF in at least one tissue through TWAS analysis of the original GWAS_{HF} data after Bonferroni correction. We also conducted S-PrediXcan on GWAS_{HF} as a comparison.

Functional characterization and contextual analysis for risk genes

To gain insight into the biological processes associated with TWAS_{MTAG} risk genes, we conducted pathway enrichment analysis using the web-based tool g:Profiler.³⁷ We utilized multiple ontology and pathway databases, including Gene Ontology, WikiPathways, Human Protein Atlas, and KEGG. Statistical significance was determined using the Benjamini-Hochberg FDR correction with a threshold of 0.05. This approach allowed us to identify pathways with biological relevance to HF. Using the same methodology, we conducted separate enrichment analyses on the TWAS_{GWAS} risk genes for comparative purposes. We also applied this analytical approach to GWAS_{CAD} genes.

We employed the scDRS method³⁸ to explore the association between cardiac-relevant cells and HF risk. We selected the top 1,000 genes ranked by p value from the TWAS analysis of MTAG_{HF} results for this analysis. ScDRS integrates single-cell RNA sequencing data with gene-disease association information derived from GWAS.³⁸ It links individual cells to disease status by assessing the excess expression of these top-ranked genes within each cell. ScDRS implements a unified Monte Carlo (MC) test for downstream analysis to evaluate the association between cell types and disease. Specifically, the top 5% quantile of the disease score from cells of a given cell type is used as the

test statistic, which is robust to annotation outliers. MC p values and Z scores are calculated for each cell type to determine statistical significance and further prioritize associations that reached the MC limit. The MC p values are computed by comparing the actual test statistics with control scores, while MC Z scores provide assessment unaffected by MC limitations through standardization. We also applied the scDRS method to the top 1,000 genes ranked by p value from TWAS analysis performed on GWAS_{HF} results for comparative analysis.

Furthermore, we investigated the functional enrichment of protein-protein interaction (PPI) networks among risk genes identified from TWAS via the STRING v.12.0 database.³⁹ We then used the cytoHubba plugin in Cytoscape to determine the top 10 genes as hub genes.⁴⁰

Querying the MGI database

To confirm the roles of genes identified by MTAG and hub genes in our study within cardiovascular diseases, we queried the Mouse Genome Informatics (MGI) (<http://www.informatics.jax.org/>) resource⁴¹ for information on knockout models associated with them. MGI is a pivotal bioinformatics resource, providing comprehensive datasets of the mouse genome comprising gene functionalities, expression patterns, mutation profiles, and data related to disease models. We focused on genes for which cardiovascular abnormalities became apparent after knockout experiments.⁴¹

Drug repurposing analysis

To translate our genetic findings into therapeutic strategies for treating HF, we conducted drug repurposing analysis with the following key steps: first, we selected all TWAS_{MTAG} risk genes and MTAG_{HF} genes. Subsequently, we used the GEN2FUNC function within FUMA to map these genes onto the DrugBank database to generate a preliminary list of candidate drugs.⁴² We then obtained a list of 26 FDA-approved HF drugs from Drug Bank⁴³ (Table S2).

For comprehensive drug similarity assessment, we implemented three complementary methods inspired by Li et al.⁴⁴: (1) chemical structure-based similarity using the Tanimoto coefficient calculated from molecular fingerprints, which quantifies the structural overlap between drug compounds; (2) pathway-based functional similarity that evaluates the shared biological pathways between drug targets; and (3) target protein sequence similarity that considers evolutionary relationships between drug targets. Using DrugBank v.5.1.13 data, we calculated each similarity measure independently for every candidate drug paired with each FDA-approved HF drug. We then computed a comprehensive similarity score by averaging these three measures, providing a multi-dimensional assessment of drug similarity. Finally, we identified drugs with average comprehensive similarity scores greater than 0.2 as potential HF therapeutic candidates. As a comparison, we also performed drug repurposing using 27 reported MTAG_{HF} genes and the TWAS_{GWAS} risk genes.

Results

LDSC analysis revealed the polygenicity of HF and CAD and the strong genetic correlation between them

Single-trait LDSC analysis suggested that the inflation observed in the test statistics is likely a consequence of polygenicity, rather than population stratification (Table S3).

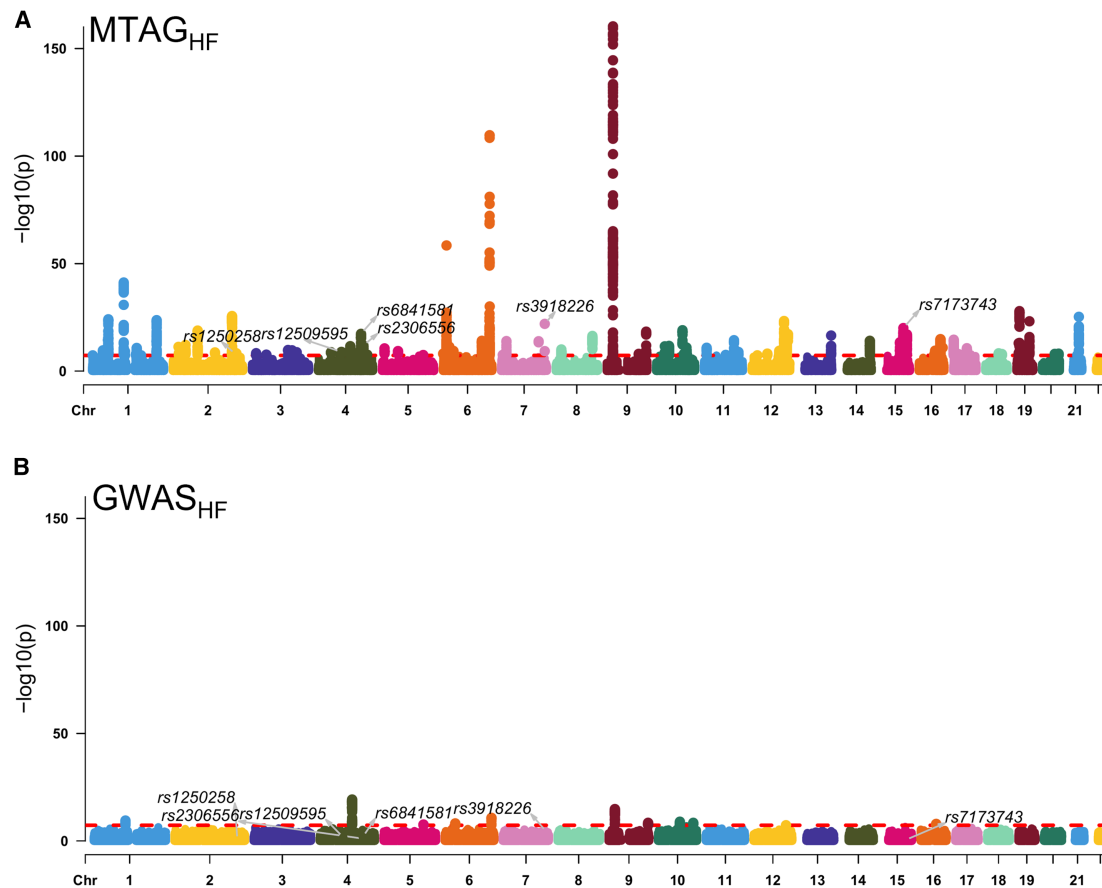


Figure 2. MTAG analysis boosted SNP discovery power for HF

Manhattan plots for $MTAG_{HF}$ (A) and $GWAS_{HF}$ (B). Horizontal red dashed lines indicate the genome-wide significance threshold ($p < 5 \times 10^{-8}$). SNPs are labeled in place on chromosome coordinates.

The mean χ^2 statistics were both greater than 1.1. LDSC intercepts consistently remained close to 1. Pairwise LDSC analysis further revealed a strong positive genetic correlation between HF and CAD ($r_g = 0.67$, $SE = 0.03$; $p = 8.03 \times 10^{-112}$). These findings supported our rationale for conducting a multi-trait analysis of HF and CAD.

MTAG analysis identified HF-associated risk loci with evidence of replication

We conducted MTAG on $GWAS_{HF}$ and $GWAS_{CAD}$, and identified 4,899 SNPs significantly associated with HF ($P_{MTAG_{HF}} < 5 \times 10^{-8}$), among which 4,749 SNPs did not reach genome-wide significance in the original $GWAS_{HF}$ analysis (Figures 2A and 2B); additionally, the Q-Q plots for the GWAS and MTAG results can be found in Figures S1A and S1B. For instance, rs6841581, located within the exon of the *EDNRA* gene, demonstrated an increased significance in MTAG analysis ($P_{MTAG_{HF}} = 4.75 \times 10^{-18}$) compared with $GWAS_{HF}$ analysis ($P_{GWAS} = 1.41 \times 10^{-2}$). Similarly, rs3918226, located within the intron of the *NOS3* (OMIM: 163729) gene, exhibited an increase in significance from $GWAS_{HF}$ analysis ($P_{GWAS} = 5.98 \times 10^{-5}$) to $MTAG_{HF}$ analysis ($P_{MTAG_{HF}} = 1.09 \times 10^{-22}$). From the original GWAS to multi-trait results, the mean χ^2 statistics increased from 1.158 to 1.340. These re-

sults suggested that MTAG analysis could be more powerful than single-trait analysis because it leverages the genetic correlation among multiple traits, and the maximum FDR for $MTAG_{HF}$ is 0.03, indicating no overall inflation due to violation of the homogeneous assumption.

Through FUMA annotation of the 4,899 significant SNPs, we pinpointed 99 risk loci from $MTAG_{HF}$ (Figure S2). Among these loci, 27 were previously reported, while 72 were identified using MTAG. We compared our $MTAG_{HF}$ loci with original $GWAS_{CAD}$ loci to evaluate overlap. Among the 99 identified loci, 46 (46.46%) overlapped with CAD-associated SNPs. Of the 72 MTAG-identified loci, 35 (48.61%) shared signals with CAD, indicating that, while CAD information contributes to power via MTAG, a substantial fraction of $MTAG_{HF}$ loci do not overlap $GWAS_{CAD}$ and may reflect HF-relevant mechanisms.

Since replication analysis is recommended to evaluate the credibility of each SNP association when applying MTAG to low-powered GWAS or GWAS characterized by significant heterogeneity in statistical power, we conducted MTAG analysis on the FinnGen cohort as a replication analysis. In the replication phase, we evaluated consistency of effect directions and nominal significance for $MTAG_{HF,R}$ rather than re-testing genome-wide significance, as the

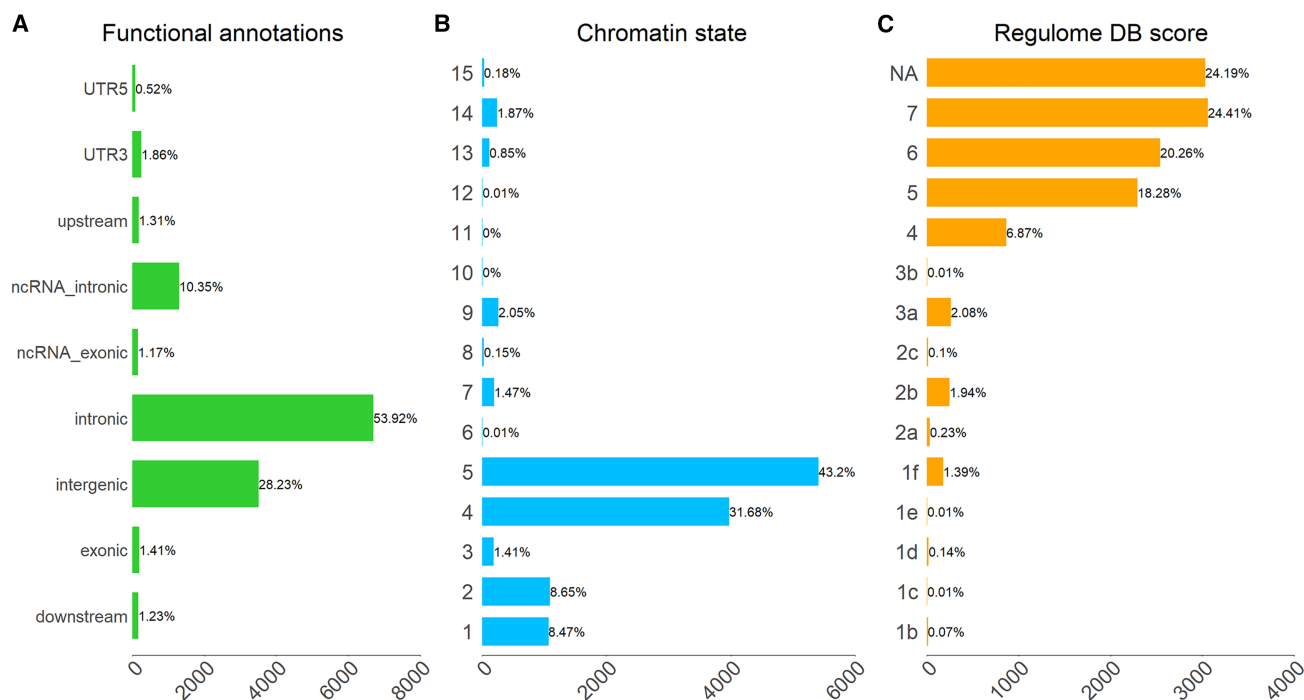


Figure 3. Functional annotations of candidate SNPs

(A) Functional annotation of candidate SNPs.

(B) Distribution of candidate SNPs across 15 categories of minimum chromatin state (chromatin states with a value ≤ 7 are considered as open chromatin regions).

(C) RegulomeDB scores of candidate SNPs. RegulomeDB scores from 1b to 2c are considered likely to affect transcription factor binding.

discovery phase ($p < 5 \times 10^{-8}$) had already identified high-confidence loci through stringent multiple testing correction. The nominal significance threshold ($p < 0.05$) combined with directional consistency is acceptable as supportive evidence.^{8,45}

Our findings revealed that, out of the 99 loci, 82 exhibited supportive replication in MTAG_{HF}. Additionally, among the 72 MTAG-identified loci identified, 58 were supported in replication (Table S4). Notable examples included the previously mentioned rs6841581 in *EDNRA* ($P_{\text{MTAG}_{\text{HF}}_{\text{R}}} = 2.63 \times 10^{-16}$) and rs3918226 in *NOS3* ($P_{\text{MTAG}_{\text{HF}}_{\text{R}}} = 1.30 \times 10^{-8}$). Additionally, other associations such as rs7173743 in *MORF4L1* ($P_{\text{MTAG}_{\text{HF}}_{\text{R}}} = 7.17 \times 10^{-12}$, OMIM: 607303), rs12509595 in *FGF5* ($P_{\text{MTAG}_{\text{HF}}_{\text{R}}} = 4.81 \times 10^{-3}$, OMIM: 165190), rs2306556 in *GUCY1A3* ($P_{\text{MTAG}_{\text{HF}}_{\text{R}}} = 5.43 \times 10^{-4}$, OMIM: 139396), and rs1250258 in *FN1* ($P_{\text{MTAG}_{\text{HF}}_{\text{R}}} = 2.04 \times 10^{-3}$, OMIM: 135600) were also successfully replicated.

Functional annotation for significant SNPs of MTAG_{HF}

We conducted FUMA to investigate the functional characteristics of significant SNPs (Table S5). For 12,410 candidate SNPs within risk loci, most (97.42%) were in non-coding regions, including UTR3, UTR5, downstream, intergenic region, intron, ncRNA intron, and upstream. Only a small proportion of SNPs were exonic, with 175 (1.41%) in coding RNA and 145 (1.17%) in non-coding RNA (Figure 3A). Among exonic SNPs in coding

RNA, rs10965215 was the most statistically significant ($P_{\text{MTAG}_{\text{HF}}} = 2.87 \times 10^{-58}$) and was mapped to the *CDKN2B-AS1* (OMIM: 613149) gene. This was followed by rs3798220 ($P_{\text{MTAG}_{\text{HF}}} = 6.74 \times 10^{-51}$), which was mapped to the *LPA* (OMIM: 152200) gene. The most significant exonic SNP was rs564398 in a non-coding RNA ($P_{\text{MTAG}_{\text{HF}}} = 9.52 \times 10^{-66}$), mapped to the *CDKN2B-AS1* gene.

Furthermore, 94.89% of candidate SNPs were located in open chromatin regions (minimum chromatin state ≤ 7) (Figure 3B); 3.89% of candidate SNPs were likely to affect the binding of transcription factors (RegulomeDB scores from 1b to 2c) (Figure 3C); and 3.62% of candidate SNPs were deemed potentially deleterious (CADD > 12.37). Notably, the SNP with the highest CADD score was rs328 (CADD score 50) located in the 8p21.3 locus. It is an exonic SNP of gene *LPL* (OMIM: 246650). All genes mentioned above (*LPA*, *CDKN2B-AS1*, and *LPL*) have been associated with HF⁴⁶ and CAD^{47,48} in previous studies, now with a higher level of evidence offered in this study.

Defining SNP credible sets within risk loci

To facilitate the shortlisting of causal SNPs for each of the 99 MTAG_{HF} risk loci, we identified 1,982 SNPs within the 90% credible set using Bayesian fine mapping test (see materials and methods and Table S6). The credible set of 10 HF loci contained only the top SNP with posterior probability greater than 0.90. The credible set of 85 HF

loci contained multiple SNPs. The credible set of the remaining four risk loci (1p36.32, 4q33, 5q31.3, 21q22.12) contained no SNPs.

Two of the 10 loci with exactly one top SNP were particularly noticeable. One was the top SNP rs3918226 ($P_{\text{MTAG_HF}} = 1.09 \times 10^{-22}$) in locus 7q36.1. This SNP was associated with systolic blood pressure⁴⁹ and myocardial infarction,⁵⁰ and it was an intron of the *NOS3* gene, with a CADD score of 12.89. The protein encoded by the *NOS3* gene synthesizes endothelial nitric oxide, modulates vascular function, and maintains cardiovascular health. Research has also revealed that mice with a knockout of the *NOS3* gene exhibit cardiovascular abnormalities, including increased heart weight and abnormal cardiac muscle relaxation.⁵¹

The other top SNP was rs7412 ($P_{\text{MTAG_HF}} = 6.95 \times 10^{-24}$) in locus 19q13.32. It was in an exon of the *APOE* (OMIM: 107741) gene and possessed a high CADD score of 25.1. Additionally, it has been associated with low-density lipoprotein cholesterol measurement,⁵² triglycerides,⁵³ and pulse pressure.⁴⁹ The *APOE* gene encodes a protein called apolipoprotein E, which plays a crucial role in lipid metabolism, including cholesterol transport and lipoprotein clearance, influencing cardiovascular health. A study has indicated that the knockout of *APOE* in mice increases the risk of atherosclerosis in arteries.⁵⁴

Functional enrichment of significant MTAG_{HF} SNPs

We observed 780 significant enrichments of significant MTAG_{HF} SNPs in regulatory and functional categories (Table S7). Among the significant SNPs of GWAS_{HF} results, only 7 reached the significance threshold (Table S8). The most significant enrichment of SNPs was found in exons of genic regions ($\text{OR} = 3.76$, $p = 2.34 \times 10^{-10}$) (Figure 4A). In terms of tissue-specific DNase I hypersensitive sites (Figure 4B), SNPs were significantly enriched in tissues, such as fetal heart, blood vessel, and embryonic stem cell, with fetal heart being the most significant ($\text{OR} = 4.94$, $p = 6.14 \times 10^{-18}$). Concerning tissue-specific chromatin states (Figure 4C), SNPs were significantly enriched in the transcribed region, with the most significant enrichment found in the actively transcribed region of liver tissue ($\text{OR} = 2.67$, $p = 1.00 \times 10^{-8}$). In terms of tissue-specific histone-modified regions, SNPs showed a significant enrichment across tissues, such as blood and liver (Figure 4D), with the H3K4me1 region of blood tissues ($\text{OR} = 3.22$, $p = 2.45 \times 10^{-11}$) being the most significant.

Uncovering HF risk genes through TWAS

To uncover HF risk genes for subsequent in-depth functional analysis, we conducted a comprehensive TWAS on MTAG_{HF} (see materials and methods). After Bonferroni correction to adjust for multiple testing within each tissue, we identified 215 TWAS_{MTAG} risk genes that displayed significant associations with HF in at least one tissue type (Table S9).

Among these TWAS_{MTAG} risk genes, two were noteworthy. The first was *EDNRA* ($P_{\text{TWAS}} = 1.72 \times 10^{-17}$ in adipose visceral omentum tissue), which was previously linked to ischemic stroke⁵⁵ and pulse pressure,⁵⁶ but was associated with HF in this study. Importantly, it was absent from the TWAS_{GWAS} risk genes (Table S10). *EDNRA* encodes the endothelin receptor type A, a G-protein-coupled receptor predominantly expressed in endothelial and smooth muscle cells. It plays a pivotal role in mediating the signaling of the endothelin-1 hormone. Research suggested that the deficiency in this gene could result in cardiovascular anomalies in mice, such as elevated blood pressure, abnormal cardiovascular system morphology, and right ventricle dilation.⁵⁷

The second gene of significance is *FURIN* ($P_{\text{TWAS}} = 3.09 \times 10^{-15}$ in liver tissue), a gene encoding a protein known as furin, in the family of serine proteases. This protease primarily functions within the cells' Golgi apparatus and vesicles, playing a pivotal role in the post-translational modification and processing of proteins. Research indicated that the absence of furin leads to severe cardiovascular anomalies, including anomalous cardinal vein morphology, the absence of vitelline blood vessels, and abnormal heart development in developing mouse embryos.⁵⁸

Contextual analysis of TWAS risk genes

To assess the relative specificity of these pathways to HF versus CAD, we conducted a comparative pathway analysis between TWAS_{MTAG} risk genes and GWAS_{CAD} genes. The analysis revealed that CAD-associated genes were significantly enriched in 1,065 pathways (Table S11), while our TWAS_{MTAG} risk genes were enriched in 414 pathways (Table S12). Among these pathways, 273 were shared between the 2 diseases, whereas 141 pathways (34.0%) were enriched only among the TWAS_{MTAG} risk genes, highlighting biological processes that appear more HF enriched in our analysis. These HF-enriched pathways included myofibroblast differentiation (adjusted $p = 4.26 \times 10^{-2}$), regulation of muscle adaptation (adjusted $p = 4.96 \times 10^{-2}$), and heart muscle; cardiomyocytes[≥medium] (adjusted $p = 8.09 \times 10^{-3}$). Meanwhile, pathways shared with CAD included response to endogenous stimulus (adjusted $p = 1.78 \times 10^{-3}$), cell population proliferation (adjusted $p = 9.21 \times 10^{-3}$), and phosphate-containing compound metabolic process (adjusted $p = 1.12 \times 10^{-2}$). Based on these findings, we selected 13 representative pathways with biological relevance for presentation (Figure 5A), of which 10 were enriched only among the TWAS_{MTAG} risk genes and 3 were shared with CAD. This partial overlap suggests that, while our MTAG approach leveraged CAD data to enhance statistical power, it also successfully identified biological processes that are relatively more associated with HF pathophysiology in our data. In contrast, TWAS_{GWAS} risk genes were enriched in only five pathways: Golgi stack, Golgi membrane, Golgi cisterna, I band, and Z disc (Table S13).

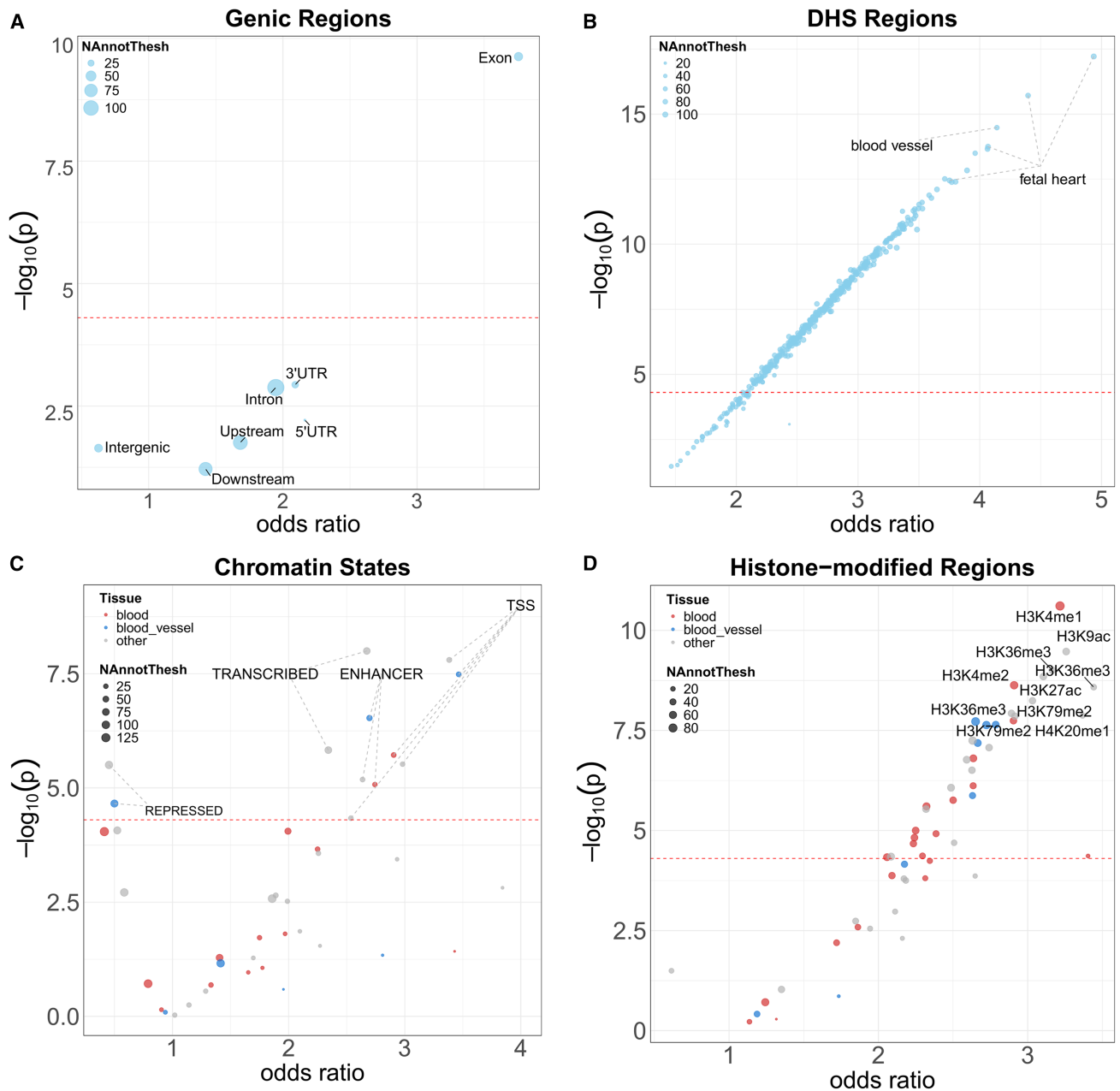


Figure 4. Functional enrichment of significant MTAG_{HF} SNPs

A total of 4,899 SNPs were analyzed with GARFIELD. Enrichment by (A) genic regions, (B) tissue-specific DNase I hypersensitive site (DHS) regions, (C) tissue-specific chromatin states, and (D) tissue-specific histone-modified regions. In all plots, enrichment odds ratio were plotted against $-\log_{10}(p)$ value. The dotted horizontal lines mark the statistical significance threshold of $p = 0.05/1005$. Size of dot markers (NAnnotThesh) is proportional to the number of independent significant SNPs in the specific annotation category, while in subplots (C) and (D), markers are also color-coded by tissue categories.

Using single-cell enrichment analysis with the top 1,000 genes ranked by their p values from our TWAS analysis of MTAG_{HF} results, we found that they were highly enriched in smooth muscle cells ($p = 4.00 \times 10^{-3}$), fibroblasts of cardiac tissue ($p = 1.40 \times 10^{-2}$), and cardiac endothelial cells ($p = 1.80 \times 10^{-2}$) (Figure 5B; Table S14). In contrast, the top 1,000 genes from TWAS analysis of GWAS_{HF} showed no enrichment in any cardiac cells (Table S15), further demonstrating the enhanced sensitivity of our multi-trait approach in

identifying cell-type-specific associations relevant to HF pathophysiology.

We also constructed a protein-PPI network on the TWAS_{MTAG} risk genes, resulting in a network with 204 nodes and 402 edges (Figure S3). The PPI network demonstrated significant enrichment ($p < 1 \times 10^{-16}$), indicating that the proteins in this network are biologically interconnected. Among all genes, 10 hub genes were identified: *APOE*, *APOB* (OMIM: 107730), *PCSK9* (OMIM: 607786), *LPL*, *SCARB1* (OMIM: 601040), *LIPC* (OMIM: 151670),

Table 1. Fourteen FDA-approved drugs were identified as candidate drugs for HF

DrugBank ID	Generic name	Target	Similarity score
DB00559	bosentan	<i>EDNRA</i>	0.4589
DB00277	theophylline	<i>PDE5A</i>	0.4084
DB06237	avanafil	<i>PDE5A</i>	0.3261
DB06267	udenafil	<i>PDE5A</i>	0.2772
DB04951	pirfenidone	<i>FURIN</i>	0.2707
DB01125	anisindione	<i>GGCX</i>	0.2569
DB09237	levamlodipine	<i>NOS3</i>	0.2422
DB06403	ambrisentan	<i>EDNRA</i>	0.2416
DB06268	sitaxentan	<i>EDNRA</i>	0.2319
DB08908	dimethyl fumarate	<i>KEAP1</i>	0.2198
DB01022	vitamin K1	<i>GGCX</i>	0.2143
DB03756	doconexent	<i>SREBF1</i>	0.2111
DB01162	terazosin	<i>TGFB1</i>	0.2093
DB08932	macitentan	<i>EDNRA</i>	0.2061

Drug repurposing analysis

We subsequently identified 81 potential therapeutic drugs based on MTAG_{HF} and TWAS_{MTAG} risk genes (see [materials and methods](#) and [Table S17](#)), 14 of which were approved by the US FDA (see [Table 1](#)) for clinical use. Notably, 4 of them (DB06403, DB00559, DB08932, and DB06268) have undergone clinical trials for the treatment indication for HF.^{59–62} Another one (DB09237) has been subjected to a clinical trial for curing hypertension which is closely related to HF. There are an additional 67 candidate drugs that are investigational compounds that are yet to be approved by the US FDA for the treatment of any disease. In contrast, using the 27 previously reported GWAS_{HF} genes and the TWAS_{GWAS} risk genes, we could only identify 10 drug compounds. Among these, only one compound had an overall similarity score to FDA-approved HF drugs exceeding 0.20. In contrast, the remaining nine compounds all had similarity scores below the threshold required for repurposing ([Table S18](#)).

Discussion

In this study, we conducted a multi-trait analysis by combining HF and CAD disease cohorts, identifying 72 HF-associated loci from MTAG_{HF} and 215 HF risk genes through TWAS. Based on these genetic discoveries, we conducted drug repurposing and identified 81 potential therapeutic drug compounds for HF, including 14 US FDA-approved drugs.

A significant challenge in HF genetic research is its moderate heritability (approximately 26%⁶) compared with other cardiovascular traits like CAD. This limited genetic contribution, combined with HF's complex nature as a

clinical endpoint representing multiple pathophysiological pathways, results in a dispersed genetic architecture with smaller effect sizes for individual variants. Our multi-trait approach effectively overcomes some of these challenges by leveraging the strong genetic correlation between HF and CAD ($r_g = 0.67$), which was confirmed using summary statistics from the largest available HF and CAD GWAS datasets. This strategy was particularly promising given the dual relationship between HF and CAD. They are not only genetically correlated but also causally related, with CAD being a major etiological factor for HF development through myocardial infarction leading to cardiac remodeling. While this dual relationship enhanced our statistical power to identify genetic variants influencing HF risk, including those acting through CAD-related mechanisms (48.61% of MTAG-identified loci overlapped with CAD signals), it also presented a challenge in identifying HF-related genetic pathways. Nevertheless, our approach successfully revealed that over half (51.39%) of MTAG-identified loci did not overlap with CAD, suggesting that our multi-trait analysis not only captured shared genetic architecture but also uncovered genetic factors relevant to HF that might otherwise remain undetected by conventional analyses.

We conducted MTAG analysis on the FinnGen cohort for replication of MTAG-derived HF associations. Among our 99 total identified loci, 82 exhibited supportive replications in MTAG_{HF_R}, including 58 of the 72 MTAG-identified loci, supporting the generalizability of our discoveries. Several genes mapped to these replicated loci have established cardiovascular relevance in mouse knockout models, including *EDNRA*, *NOS3*, *MORF4L1*, *FGF5*, *GUCY1A3*, and *FN1*. Among the notable risk loci, 2p21 harbors the top lead SNP rs4245791 reaching significance in

GWAS_{CAD}²⁷ and demonstrating a consistent effect direction with MTAG_{HF}. The locus mapped to the gene *ABCG8*, also known as ATP-binding cassette subfamily G member 8. *ABCG8* encodes a protein crucially involved in cholesterol transport and homeostasis. Research revealed that its deficiency in mice leads to abnormalities in heart left ventricle morphology and an increase in cardiac muscle contractility.⁶³ Another noteworthy risk locus is 15q25.1, with the top lead SNP being rs7173743. The SNP was an intronic variant of the gene *MORF4L1*. *MORF4L1*, also known as mortality factor 4-like 1, is a gene responsible for encoding a protein engaged in cellular processes, such as chromatin remodeling and DNA repair. Research demonstrated that its deficiency in mice resulted in cardiovascular abnormalities, including myocardial fiber disarray and cardiac hypertrophy.⁶⁴ We further conducted a TWAS analysis and identified 215 HF risk genes and validated the roles of these genes in the progression of HF. Pathway enrichment analysis identified several key pathways closely associated with HF, including response to endogenous stimulus, phosphate-containing compound metabolic process, and cell population proliferation. The disruption of these pathways is implicated in HF with extensive supporting evidence.

First, the response to endogenous stimulus refers to any process that induces a modification in the state or function of a cell or organism, encompassing alterations in movement, secretion, enzyme production, gene expression, and other physiological activities. Notably, the close association of endogenous hemodynamic⁶⁵ and endogenous nitric oxide⁶⁶ with HF has been widely reported.

Second, the phosphate-containing compound metabolic process pathway encompasses chemical reactions related to the phosphate group, which is the anion or salt derived from any phosphoric acid. This includes processes such as the synthesis of adenosine triphosphate. Notably, it was established that a reduction in the energy reserve available for adenosine triphosphate synthesis can render the heart more susceptible to systolic and diastolic failure.⁶⁷

Third, the cell population proliferation pathway refers to the multiplication or reproduction of cells, leading to the expansion of the cell population. The proliferation of cardiac myocytes is intricately linked to the development of HF,^{68–70} and contemporary research has proposed various approaches aimed at inducing cardiac myocyte proliferation as a therapeutic strategy for HF.^{71–73}

Furthermore, comparative pathway analysis between TWAS_{MTAG} genes and GWAS_{CAD} mapped genes revealed that, while many pathways are shared between the conditions, 34% (141 out of 414) appeared enriched among the TWAS_{MTAG} genes but not among the GWAS_{CAD} set, consistent with the identification of biological processes that are more HF-relevant in our analysis. Notably, these HF-enriched pathways included myofibroblast differentiation, regulation of muscle adaptation, and heart muscle; cardiomyocytes[medium], representing processes central to cardiac remodeling and contractile dysfunction in HF.

The enrichment in myofibroblast differentiation is particularly significant, as cardiac fibroblast-to-myofibroblast conversion represents a critical event in pathological cardiac remodeling and fibrosis.⁷⁴ Similarly, the muscle adaptation regulatory pathways highlight the importance of cardiac tissue plasticity in response to hemodynamic stress.⁷⁵ Cardiomyocyte-specific enrichment underscores the fundamental role of these cells in HF progression beyond coronary artery pathology. These unique biological signatures thus represent potential targets for therapeutic interventions in HF.

In our single-cell enrichment analysis, it is noteworthy that risk genes demonstrated enrichment in cell types, such as smooth muscle cells, fibroblasts of cardiac tissue, and cardiac endothelial cells. Additionally, an earlier study has shown that the deletion of the mineralocorticoid receptor specifically from smooth muscle cells can mitigate HF induced by transverse aortic constriction.⁷⁶ Research also indicated that endothelial cells play a crucial role in both coronary microvascular dysfunction and cardiac remodeling, ultimately contributing to the development of HF.⁷⁷

Moreover, investigations have highlighted the involvement of cardiac fibroblasts in myocardial stiffening through collagen production, as well as their active participation in modulating cardiac inflammation by synthesizing chemoattractive substances.⁷⁸ Notably, this cell-specific distribution pattern differs substantially from that observed in CAD, where genetic associations primarily implicate vascular endothelial cells and foam cell macrophages in atherosclerotic plaque formation.⁷⁹

Importantly, the enrichment of HF risk genes in cardiac-specific cell populations rather than atherosclerosis-related cell types further supports the identification of pathways specifically relevant to myocardial dysfunction and remodeling in HF, independent of coronary atherosclerosis. Interestingly, TWAS_{GWAS} risk genes showed no significant enrichment in any cardiac cell types, highlighting the enhanced biological relevance of our multi-trait approach. These findings not only confirm the biological validity of our genetic discoveries but also provide insights into potential cellular targets for therapeutic interventions in HF that might remain undetected in CAD-focused analyses.

Finally, our study leveraged genetic discoveries to identify potential therapeutic interventions through drug repurposing analysis. From MTAG_{HF} genes and TWAS_{MTAG} risk genes, we identified 81 potential therapeutic drugs for HF treatment. Notably, among these FDA-approved compounds, four (DB06403, DB00559, DB08932, and DB06268) have already undergone clinical trials for HF treatment,^{59–62} supporting the credibility of our findings. Moreover, the target genes of these potential medications exhibited substantial evidence of associations with cardiovascular system abnormalities in knockout models. Notably, *ALDH2* (OMIM: 100650), *NOS3*, *MIF*, *FURIN*, and *PDGFD*, demonstrated direct associations with such

pathological features as congestive HF, myocardial abnormalities, cardiac hypertrophy, and abnormal myocardial fiber morphology.^{51,58,80–82} Furthermore, *ALDH2*, *EDNRA*, and *FURIN*, which exhibited expression levels positively correlated with HF risk, may represent more favorable targets for the development of therapies, since therapeutics aimed at downregulating a gene are typically easier to develop than those aimed at upregulation.

In contrast, none of these FDA-approved drugs are currently approved for CAD treatment, suggesting that they may act through pathways relevant to HF pathophysiology than to coronary atherosclerosis. This finding is particularly significant as it highlights the distinct therapeutic landscape for HF compared with CAD, with drugs targeting pathways such as endothelin receptor signaling (*EDNRA* targeted by ambrisentan, bosentan, macitentan, and sitaxentan), myofibroblast activity (*FURIN* targeted by pirfenidone), and nitric oxide pathways (*NOS3* targeted by levamlodipine). These therapeutic targets align with our pathway and single-cell enrichment findings, further validating the biological relevance of our genetic discoveries. It is worth noting that, our attempts using only previously reported loci and the original GWAS_{HF} data identified just 10 drug compounds with only 1 exceeding the similarity threshold required for repurposing, highlighting the enhanced drug discovery capability of our multi-trait approach.

Despite many findings, our study had certain limitations. The GWAS data were derived from European populations, thus potentially limiting the generalizability of our study to other ethnicities. Nonetheless, recent advancements have introduced methods for transferring genetic findings from European populations to others, which may increase the utility of our research.^{83,84}

Conclusions

In summary, we identified 72 HF risk loci using MTAG with CAD. We employed the replication phase, Bayesian fine mapping, GARFIELD, and MGI queries to fully establish the biological and statistical credibility of these risk loci. We identified 215 significant HF risk genes through TWAS analysis. Pathway enrichment, single-cell enrichment, PPI network, and MGI queries confirmed the roles of these risk genes in the progression of HF. Based on these MTAG_{HF} genes and TWAS_{MTAG} risk genes, we identified 81 drug compounds potentially suitable for HF repurposing. These findings have lasting value for future efforts aimed toward the discovery of HF prevention and treatment regimens.

Data and code availability

- The datasets analyzed during the current study are publicly available.
- GWAS summary statistics data for the discovery phase are available from the GWAS Catalog under study accessions GWAS-catalog: GCST009541⁸⁵ and GWAS-catalog: GCST90132314⁸⁶ at <https://www.ebi.ac.uk/gwas/>.

- GWAS summary statistics data for the replication phase are available at https://storage.googleapis.com/finngen-public-data-r7/summary_stats/finngen_R7_I9_HEARTFAIL_ALLCAUSE.gz and https://storage.googleapis.com/finngen-public-data-r7/summary_stats/finngen_R7_I9_CHD.gz.¹⁸
- Data used in LDSC analysis can be obtained at <https://alkesgroup.broadinstitute.org/LDSCORE/>.²⁸
- JTI models of gene expression in eight heart tissues are available from Zenodo at <https://doi.org/10.5281/zenodo.3842289>.⁸⁷
- Single-cell data used in scDRS analysis are available from Tabula Sapiens Single-Cell Dataset at <https://doi.org/10.6084/m9.figshare.14267219.v5>.⁸⁸
- All drug information is available from DrugBank v.5 at <https://go.drugbank.com/>.⁸⁹
- The information on all mouse knockout models can be obtained from Mouse Genome Informatics resources at <https://www.informatics.jax.org/>.⁹⁰
- Results generated from this work are included in the published article or the [supplemental information](#).

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Author contributions

Z.Y., Y.C., K.N., and L.C.X. conceived and designed the study. Z.Y. and Z.L. collected the data. Z.Y. and Z.L. performed the data analysis. Z.Y., Y.Y., B.R., and M.L. conducted the interpretation of the analysis results. Z.Y., L.C.X., and K.N. wrote the manuscript. Y.C., Y.Y., M.L., H.L., and X.L. revised the manuscript during the peer review process. L.C.X. supervised the overall study. All authors read and approved the submitted version.

Declaration of interests

The authors declare no competing interests.

Supplemental information

Supplemental information can be found online at <https://doi.org/10.1016/j.xhgg.2025.100540>.

Web resources

Alkes Group, <https://alkesgroup.broadinstitute.org/>

DrugBank, <https://go.drugbank.com/>
 FinnGen GWAS summary statistics data, <https://storage.googleapis.com/finngen-public-data-r7>
 JTI models, <https://doi.org/10.5281/zenodo.3842289>
 Mouse Genome Informatics, <https://www.informatics.jax.org/>
 NHGRI-EBI GWAS Catalog, <https://www.ebi.ac.uk/gwas/>
 OMIM, <https://www.omim.org/>
 Tabula Sapiens Single-Cell Dataset, <https://doi.org/10.6084/m9.figshare.14267219.v5>

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Supplemental information

Multi-trait genome-wide analysis identified risk loci and candidate drugs for heart failure

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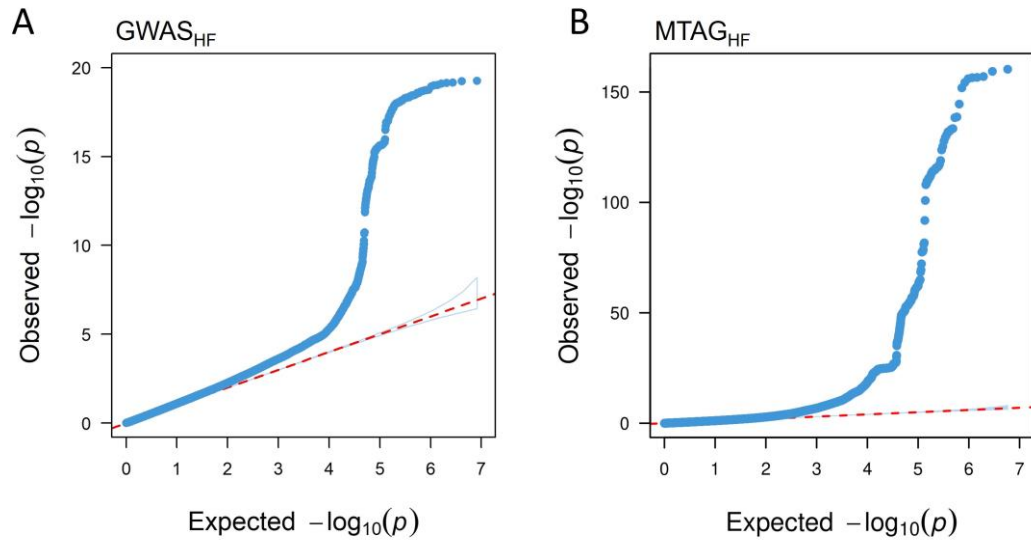


Figure S1. Q-Q plots of GWAS_{HF} and MTAG_{HF} . The Q-Q plots display the observed versus expected P value distributions for the GWAS (A) and MTAG (B) results. The blue dots represent genetic variants, the red line is the null expectation, and the gray band is its 95% confidence interval.

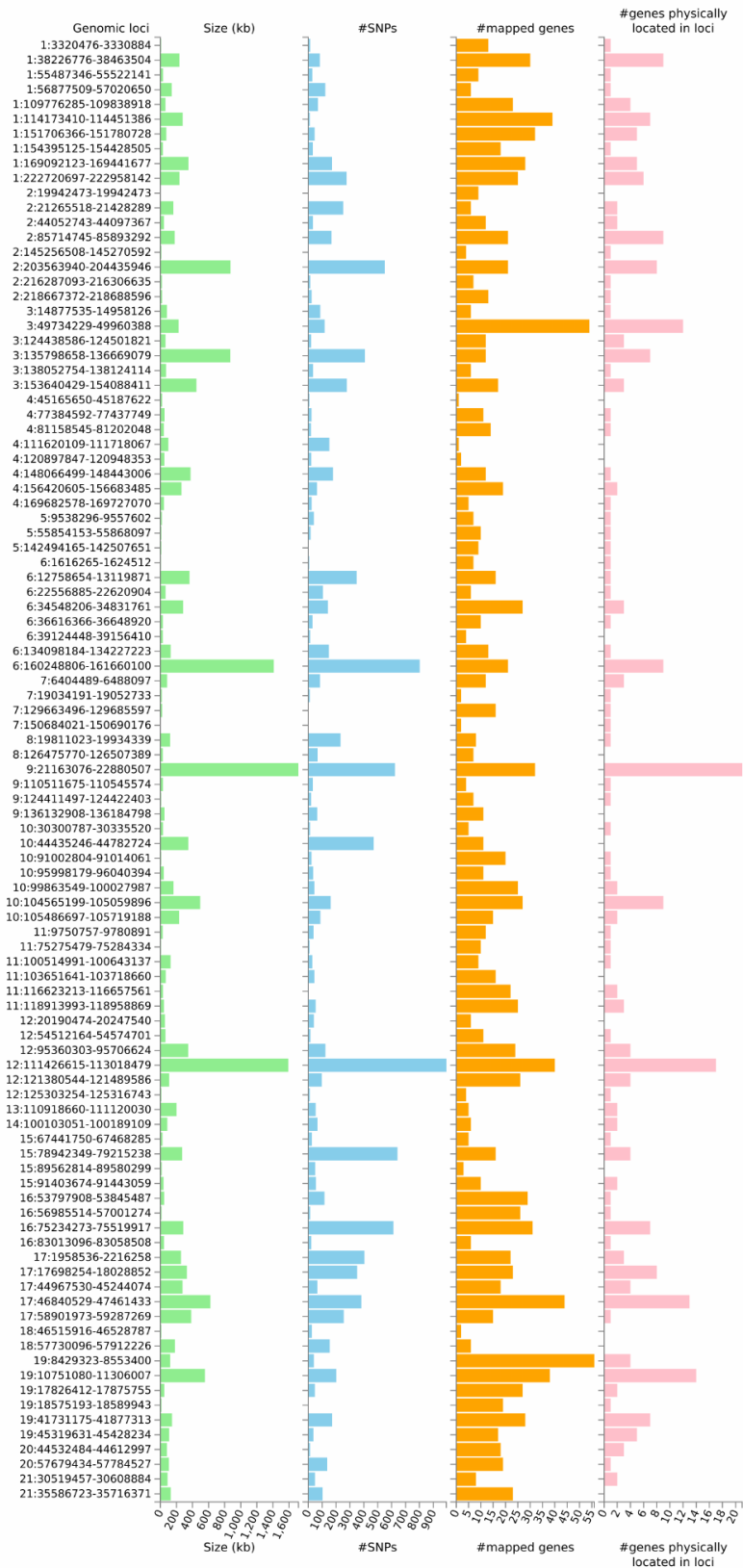


Figure S2. Summary of 99 HF risk loci. Bar graphs depicting the size of the genomic locus (left one), number of candidate SNPs in the locus (left two), number of mapped genes in the genomic locus (right two) and the number of genes physically locating within the genomic locus (right one). Candidate SNPs are in LD ($r^2=0.1$) of one of the independent significant SNPs.

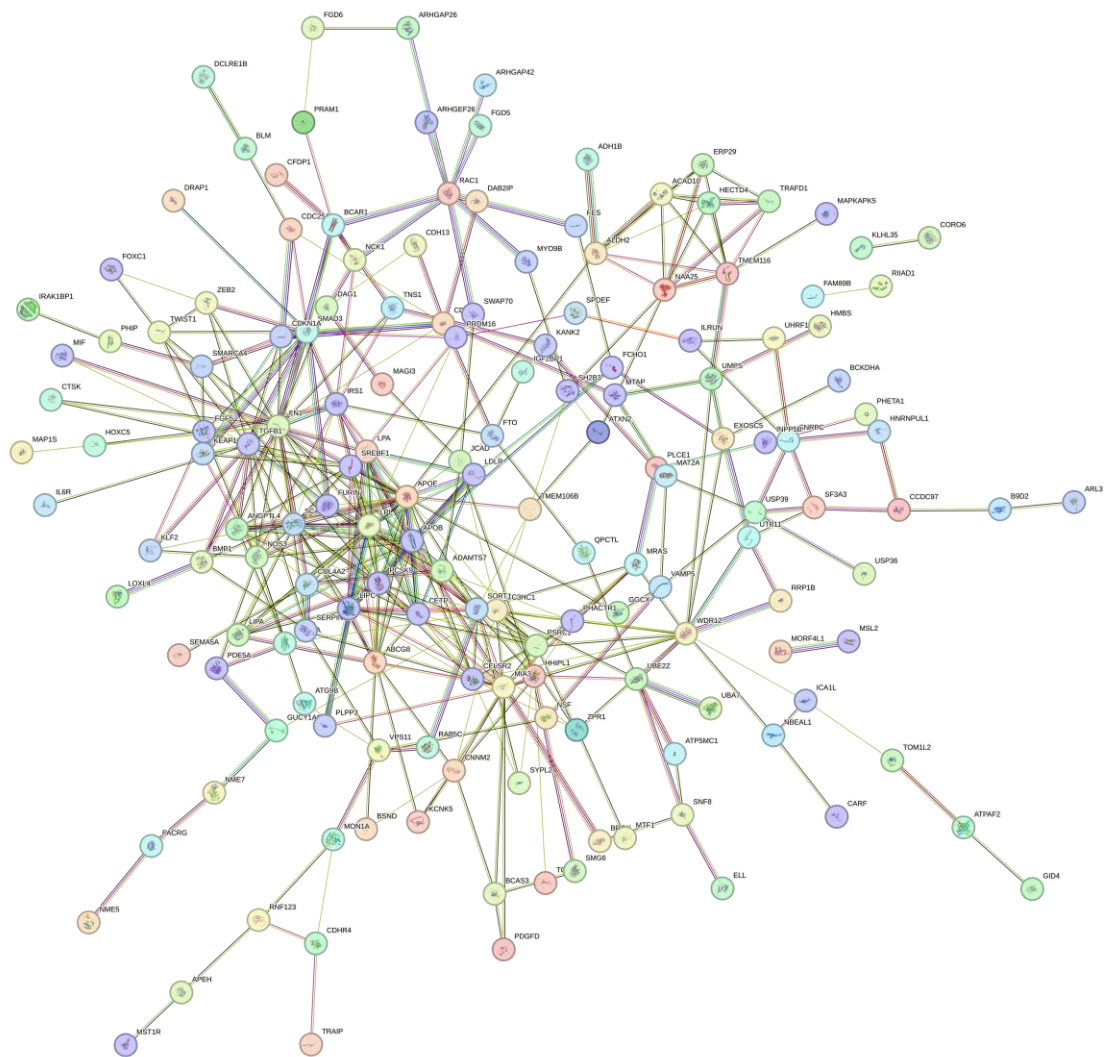


Figure S3. Protein–protein interaction network of 215 TWAS_{MTAG} risk genes identified by transcriptome-wide association study (TWAS). The reported network, including both direct (physical) and indirect (functional) associations, was based on the STRING v12.0 database. Only medium-confidence interactions between proteins are reported (interaction score > 0.4), while disconnected nodes in the network are hidden. Each node represents all the proteins produced by a single protein-coding gene locus, while edges represent protein–protein associations. Line color indicates the type of interaction evidence: light blue— from curated databases; purple—experimentally determined; green—gene neighborhood; red—gene fusions; blue—gene co-occurrence; yellow—text mining; black—co-expression; and violet—protein homology.